

Rethinking Pretraining for Specialized Design Data: Evidence from the JONES-19 Cultural Design Dataset

SUPPLEMENTARY MATERIAL

Alexandros Haridis¹ and Charles Zhou

Harvard University, Cambridge, MA 02139, USA

¹haridis@seas.harvard.edu

A.1 Hyper-Parameters table

Table A.1.1: Hyper-parameters used for training the eight baseline models on JONES-19.

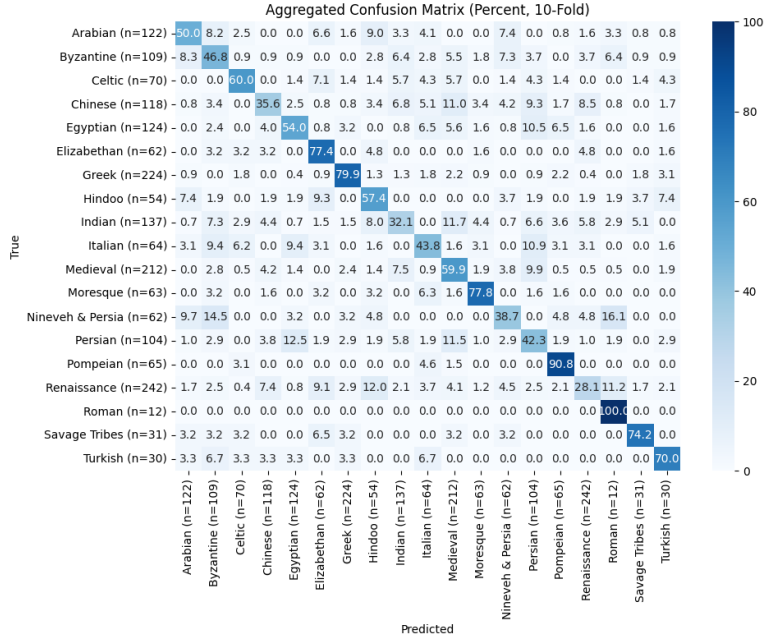
Models	Pre-training	Data Augmentation	Balancing	Resolution	Epochs	Optimizer	Loss*
ResNet18	None	Single-crop	Stratified 10-Fold	224	≤ 40	Adam	Weighted Cross Entropy
ResNet18	None	Multi-crop	Stratified 10-Fold	224	≤ 40	Adam	Weighted Cross Entropy
ResNet18	ImageNet-1k	Single-crop	Stratified 10-Fold	224	≤ 40	Adam	Weighted Cross Entropy
ResNet18	ImageNet-1k	Multi-crop	Stratified 10-Fold	224	≤ 40	Adam	Weighted Cross Entropy
ResNet50	None	Single-crop	Stratified 10-Fold	224	≤ 40	Adam	Weighted Cross Entropy
ResNet50	None	Multi-crop	Stratified 10-Fold	224	≤ 40	Adam	Weighted Cross Entropy
ResNet50	ImageNet-1k	Single-crop	Stratified 10-Fold	224	≤ 40	Adam	Weighted Cross Entropy
ResNet50	ImageNet-1k	Multi-crop	Stratified 10-Fold	224	≤ 40	Adam	Weighted Cross Entropy

* Class weights were set as the inverse of class frequencies to address label imbalance during training.

A.2 Confusion Matrices

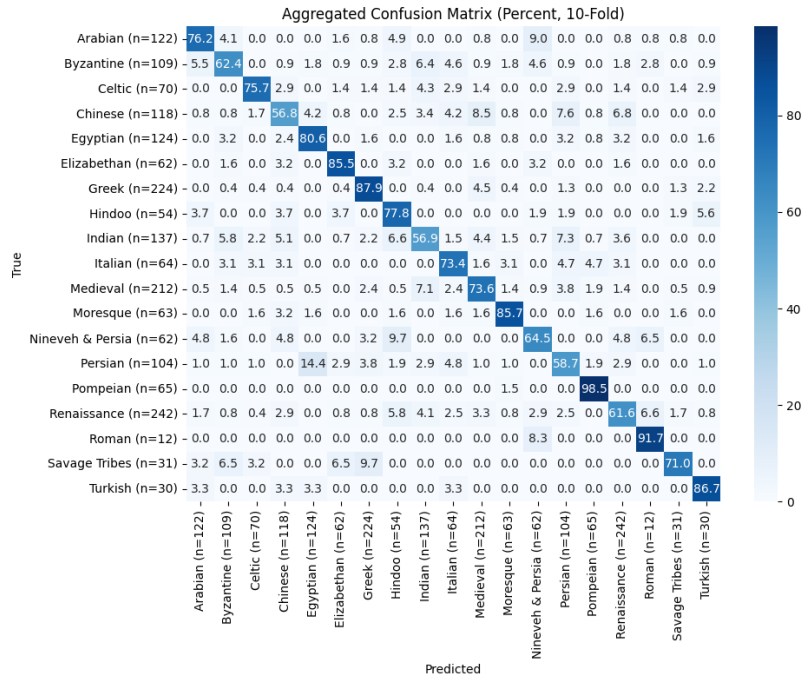
(a) ResNet18, 224 resolution, no-pretraining, singlecrop

Acc: 59.07 ± 6.71

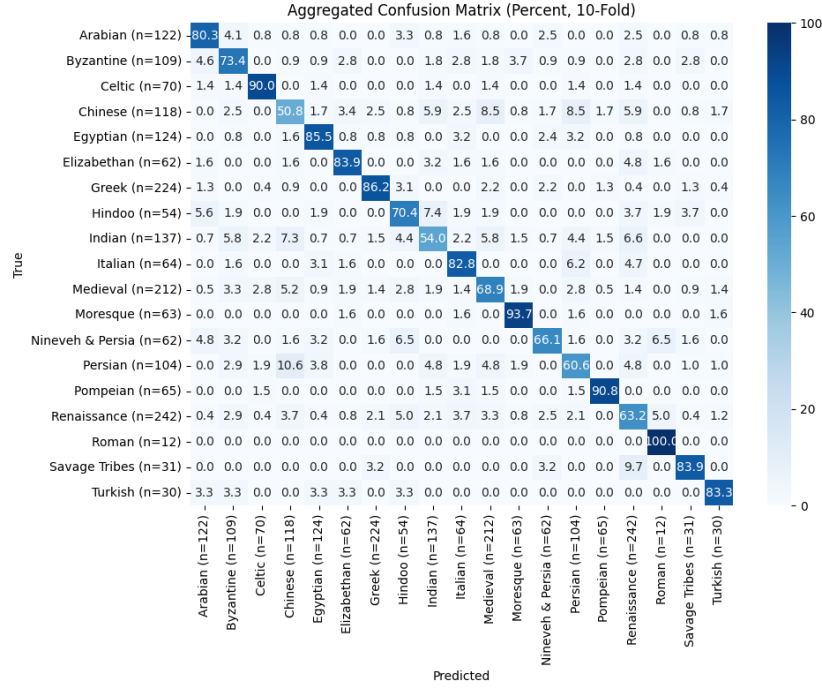


(b) ResNet18, 224 resolution, no-pretraining, multicrop

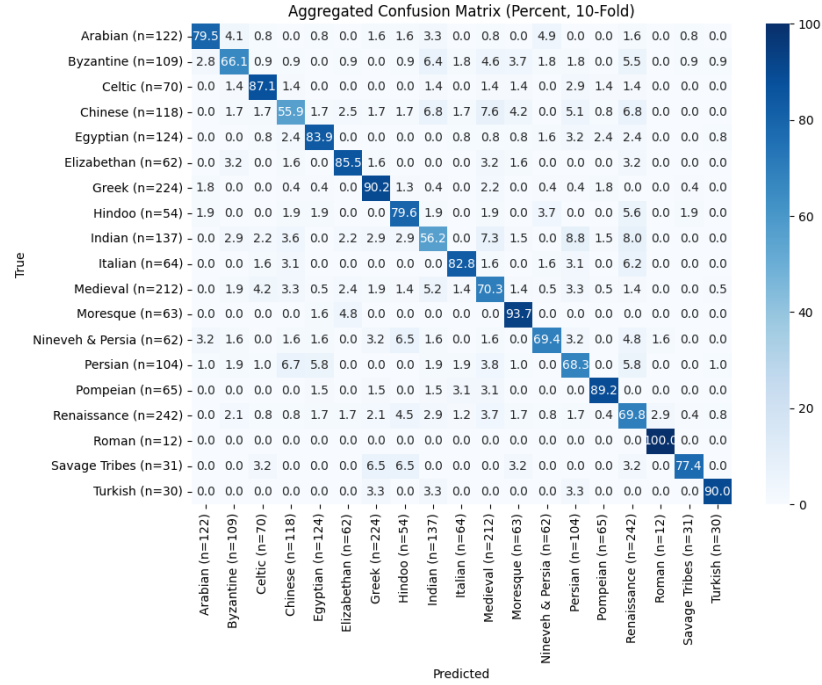
Acc: 75.31 ± 5.96



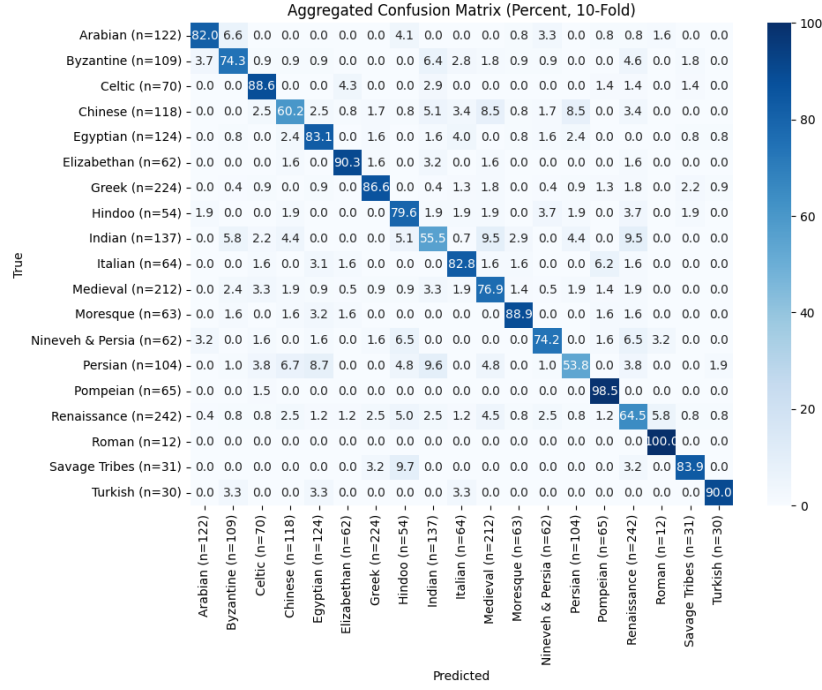
(a) ResNet18, 224 resolution, ImageNet-pretraining, singlecrop
Acc: 77.34 ± 4.03



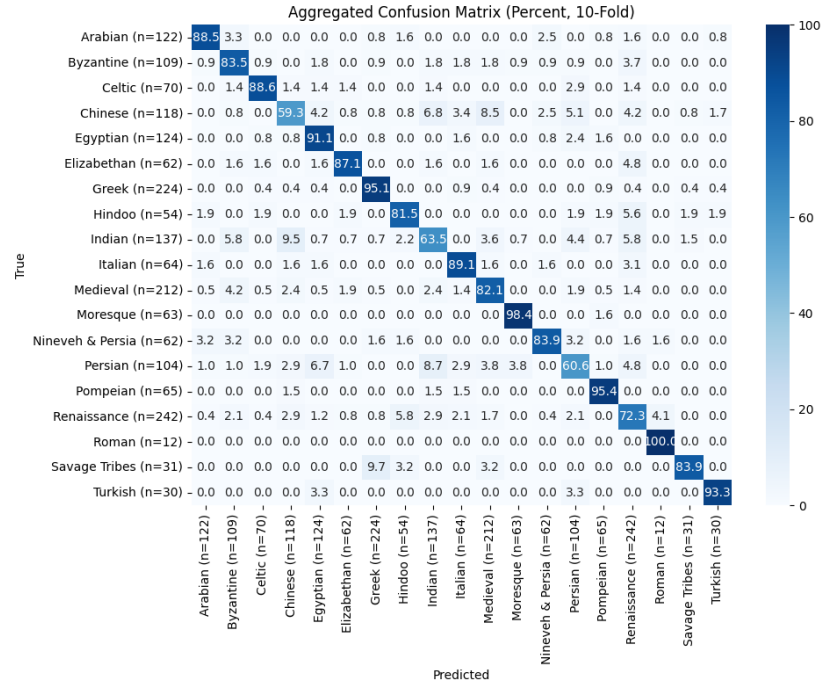
(b) ResNet18, 224 resolution, ImageNet-pretraining, multicrop
Acc: 78.76 ± 6.12



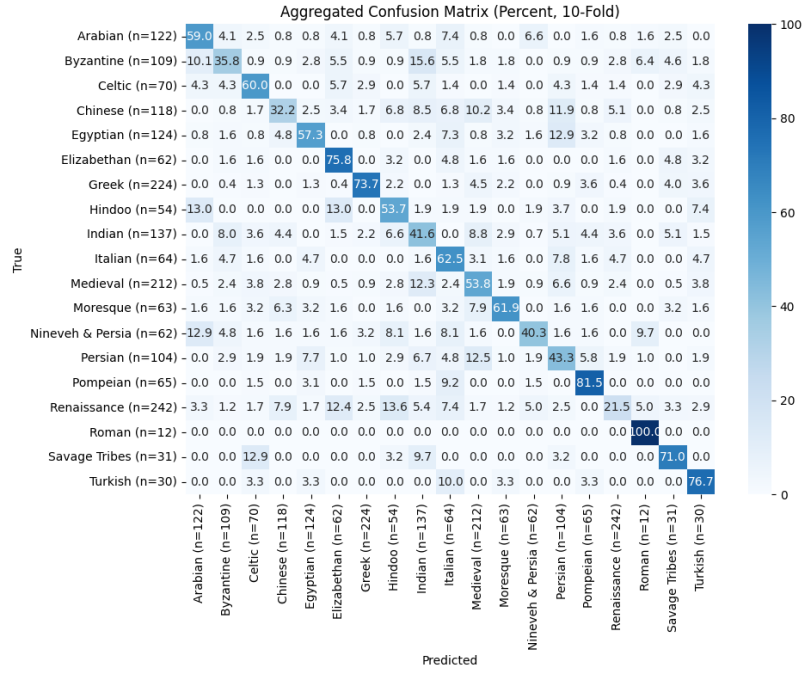
(a) ResNet50, 224 resolution, ImageNet-pretraining, singlecrop
Acc: 79.61 ± 2.85



(b) ResNet50, 224 resolution, ImageNet-pretraining, multicrop
Acc: 84.11 ± 3.95



(a) ResNet50, 224 resolution, no-pretraining, singlecrop
Acc: 58.03 ± 9.18



(b) ResNet50, 224 resolution, no-pretraining, multicrop
Acc: 77.99 ± 4.25

